



APPLICATIONS OF THE DERIVATIVE

Applications of the Derivative

- In the previous modules, you have derived and used all the rules of differentiation in finding the derivative of a function. The derivative, as you have seen in many examples, is a very powerful mathematical concept that can solve many problems involving relationships between quantities and their rates of change. In this module, you will further use this tool in the solution of certain problems. This will be divided into two sections, *optimization* and the *rate of change*.

Optimization

- In calculus, an optimization problem is a problem that involves determining optimum values. Optimum values refer to both maximum and minimum values and, more specifically, the relative extreme values of a function.

Optimization

- Optimization is a very familiar problem for many fields in human activity. In manufacturing industries, one of the most frequently considered questions among manufacturers and consumers alike, aside from the other features it offers, is the product's optimal value.

Optimization

- This makes the product more unique and affordable. In business and industry, people struggle to minimize waste and maximize productivity through their careful choices and decisions. Several methods are carried out, such as the accurate approximations to reduce unnecessary expenditures.

Optimization

- Most calculus students find solving optimization problems challenging because of certain factors. One of the challenges in solving these problems is determining the correct variables and identifying the correct quantities to optimize, either to maximize or to minimize

Optimization

- Many students wonder whether reading the answer from a calculator or from a computer generated graph can solve such problems. This is possible in some instances, but remember that functions (even the reasonably familiar ones) are hard to graph. You have to closely examine how the graph of the function behaves.

Optimization

- A quicker and yet more reliable way to solve optimization problem is by computing the derivative of the function that gives the relationship between the variables under consideration. This is because the derivative is a function of the slope of the tangent line to a curve. Also, when optimum points are in the question, the tangent line is horizontal and thus, the derivative of the function at those points is 0.

Optimization

- Some guidelines may be observed in solving optimization problems. Do not memorize them, but keep them in mind as you work through specific problems.

Optimization

- If there's a picture to draw, then draw it! Don't try to envision how things look in your head. Put a picture down on a paper and label it.
- Determine what the variables are and how they are related.
- Decide what quantity needs to be maximized or minimized.

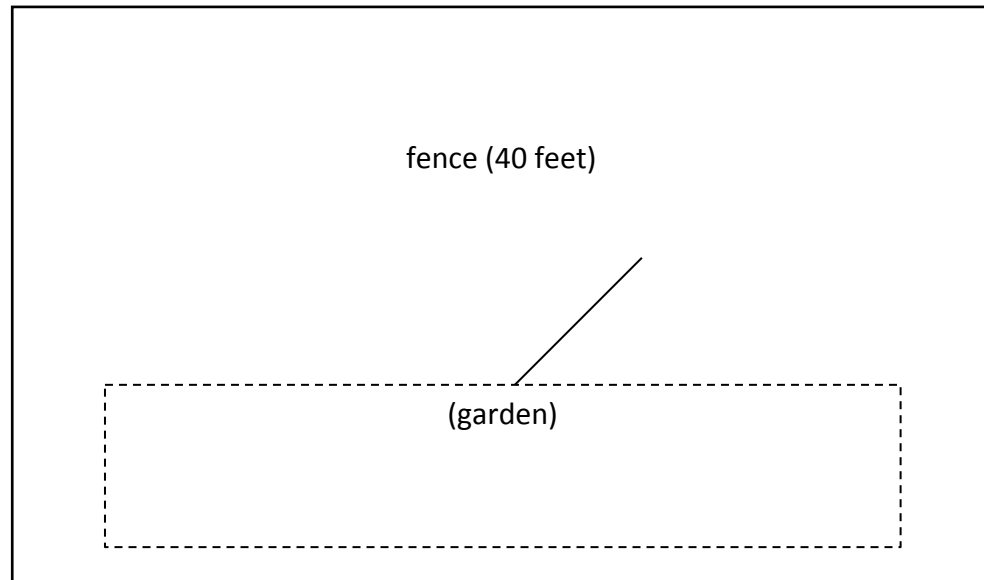
Optimization

- Write an expression for the quantity to be maximized or minimized in terms of only one variable.
- Determine the minimum and maximum allowable values (if any) of the variable you are using.
- Solve the problem. (Be sure to answer the question that is asked.)

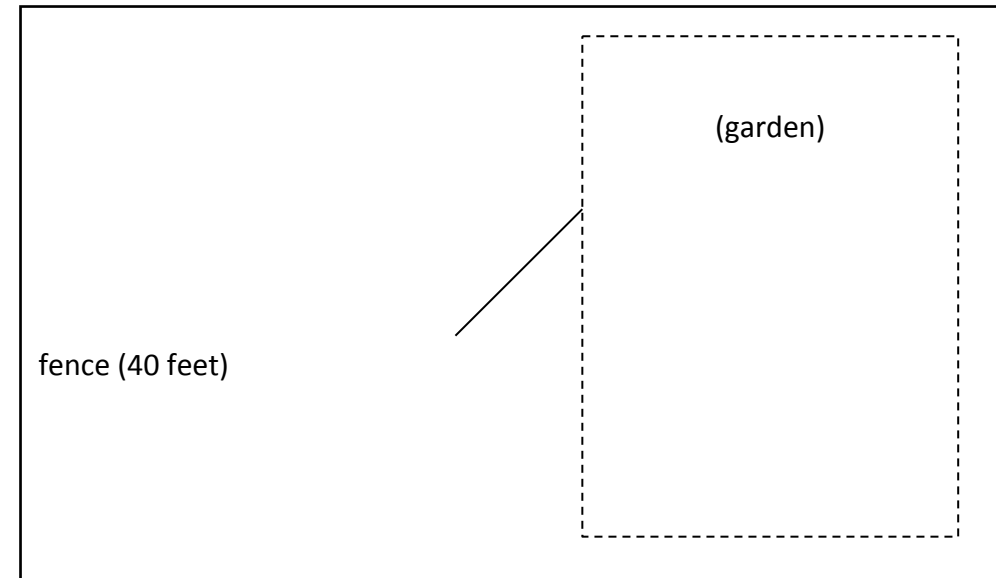
Example 1:

- You have 40 linear feet of fencing with which to enclose a rectangular space for a garden. Find the largest area that can be enclosed with this much fencing and the dimensions of the corresponding garden.

Illustrations: (Figure 12.1)



(a)



(b)

Solution:

- There are lots of possibilities. You can either enclose a plot that is very long but narrow, or enclose one that is very wide, but not very long. (See figure 12.1) How will you know which configuration is optimal? Remember that the variables for a rectangular plot are length and width, which will be represented as x and y , respectively.

Solution:

- Because you want to find the dimensions of the largest possible area, the following formula is applicable:

$$\text{Area}(A) = \text{length} \times \text{width} = xy$$

- But notice that this function contains two variables, you need to simplify it further so that the area A function is expressed as a function of only one variable. Because you have 40 linear feet of fencing, the perimeter is 40 ft.

Solution:

- You may use this to express one of the two variables in terms of the other.

$$2x + 2y = 40$$

$$2y = 40 - 2x$$

$$y = 20 - x$$

- Therefore,

$$A = x(20 - x)$$

Solution:

- Before you attempt to maximize the area $A(x)$, you need to determine if there is an interval wherein x must lie. Notice that x is a distance, so $x > 0$. Further, because the perimeter is 40, you must have $x < 20$, so the value of x must be within $0 < x < 20$.

Solution:

- By differentiation, you have

$$\begin{aligned} A' &= \frac{d}{dx} [x] \cdot (20 - x) + x \cdot \frac{d}{dx} (20 - x) \\ &= 1(20 - x) + x(-1) \\ &= 20 - x - x \\ &= 20 - 2x \end{aligned}$$

Solution:

- Now, when a quantity is at its relative maximum point, its derivative is equal to 0 at that value of x . Then,

$$A' = 20 - 2x$$

$$0 = 20 - 2x$$

$$2x = 20$$

$$x = 10$$

Solution:

- Therefore, the only critical number is $x = 10$, and this is within the interval under consideration. At $x = 10$, you have

$$A = x(20 - x)$$

$$A = 10(10) = 100$$

- Thus, the maximum area that can be enclosed with 40 feet of fencing is 100 square feet.

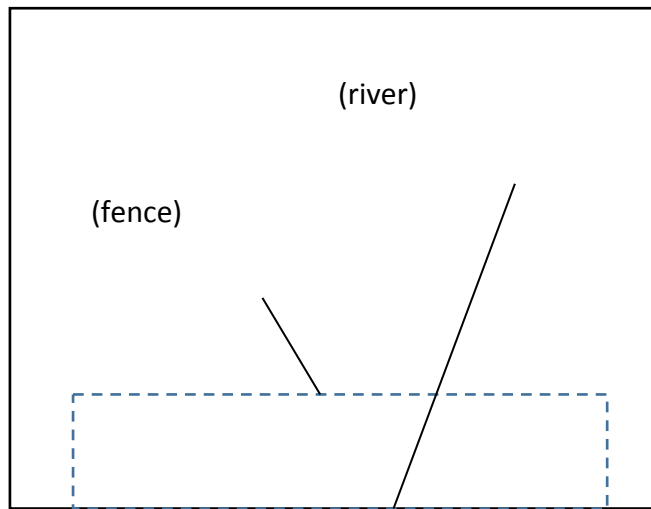
Example 2:

- A farmer has 1200 meters of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Solution:

- To get a feeling of what is happening in this problem, you may experiment with some special cases, as illustrated in figure 12.2. Note that only three sides will be fenced because the farmer needs no fence along the river.

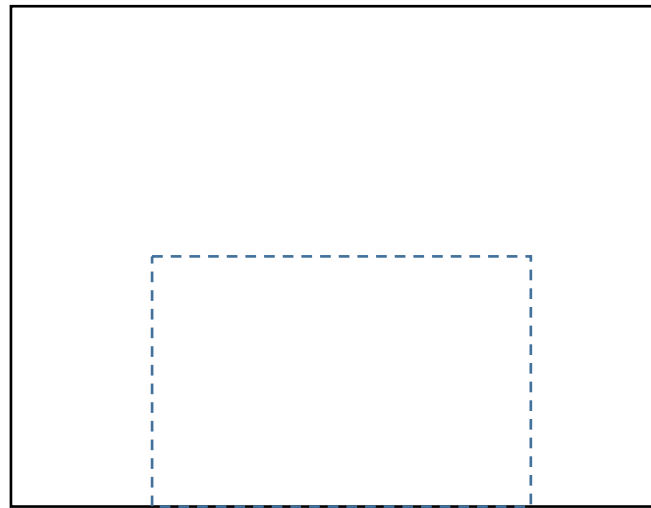
Solution: (Figure 12.2)



(a)

$$L = 1000$$

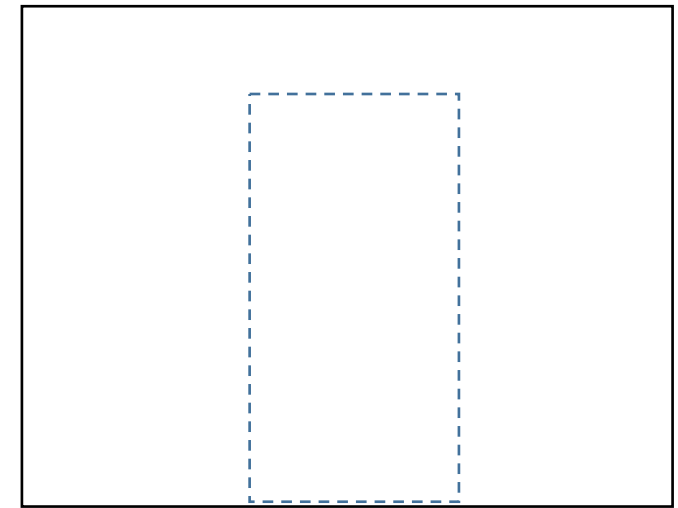
$$W = 100$$



(b)

$$L = 400$$

$$W = 400$$



(c)

$$L = 800$$

$$W = 200$$

Solution:

- You will observe a shallow, wide field, or a deep narrow field if you take figure 12.2a. A relatively small area is the result. It seems plausible that there is some intermediate configuration that produces the largest area.
- Maximizing the area A of the rectangle is shown in figure 12.2b. Let x and y be the depth and width of the rectangle (in meters).

Solution:

- Then you can express the area A in terms of x and y .

$$A = xy$$

- Expressing the area A as a function of just one variable, you can eliminate y by expressing it in terms of x . To do this, use the given information that the total length of the fencing is 1 200 meters. Thus, $2x + y = 1200$.

Solution:

- Simplifying the equation yields to $y = 1200 - 2x$, which gives

$$A = x(1200 - 2x) = 1200x - 2x^2$$

- Know that $x > 0$ and $x < 600$. Thus, the function that is maximized is

$$A(x) = 1200x - 2x^2$$

Solution:

- Then, $A'(x) = 1200 - 4x$. Setting the derivative to be 0, you have

$$A'(x) = 1200 - 4x$$

$$0 = 1200 - 4x$$

$$4x = 1200$$

$$x = 300$$

- The rectangular field should be 300 meters long and 600 meters wide.

Example 3:

- The sum of two numbers is 50. What is the smallest possible product of the two numbers?

Solution:

- Let x be the first number, and y be the second number. You want to minimize the product; thus, the working function will be $P = xy$. Because the sum of the two numbers is 50, then $y = 50 - x$, making your product function $P = x(50 - x)$. The feasible domain is the interval $0 \leq x \leq 50$.

Solution:

- Differentiating the product function, you have

$$P = x(50 - x)$$

$$\begin{aligned} P' &= x(-1) + (50 - x)(1) \\ &= -x + 50 - x \\ &= -2x + 50 \end{aligned}$$

Solution:

- Setting $P' = 0$, you have

$$P' = -2x + 50 = 0$$

$$-2x + 50 = 0$$

$$-2x = -50$$

$$x = 25$$

- Thus, $x = 25$ and $y = 25$. The maximum product of the two numbers is 625.

Applications of the Derivative

- In some cases, the function for the variable being maximized is already given so you can solve problem easily, as in the next example.

Example 4:

- A company has identified that the revenue R for a particular product is given by the equation $f(x) = -x^3 + 300x^2 + 37500x$, where x is the number of units sold. How many units need to be sold to obtain the maximum revenue?

Solution:

- Getting the derivative of the function, you have

$$\begin{aligned}f(x) &= -x^3 + 300x^2 + 37500x \\f'(x) &= -3x^2 + 600x + 37500 \\&= -3(x^2 - 200x - 12500) \\&= -3(x - 250)(x + 50)\end{aligned}$$

Solution:

- Now, recall that the maximum value occurs when $f'(x) = 0$. Thus,

$$f'(x) = -3(x - 250)(x + 50) = 0$$
$$x = 250, x = -50$$

- Because $x = -50$ is not feasible value of x then 250 units of the product will corresponds to a maximum revenue.

Applications of the Derivative

- In physics, one important application of the derivative involves the height of a free-falling object, which can be modelled using the position function

$$h = -16t^2 + v_0t + h_0$$

- where h is the height off the object in feet, t is the time in seconds, v_0 is the velocity (in feet per second), and h_0 is the initial height.

Example 5:

- A student throws a ball upward from a height of 48 feet, initially at 32 feet per second.
 - a. Write the position function that models the height of the ball t seconds after it was thrown.
 - b. What is the maximum height of the ball?

Solution:

- Because $h_0 = 48$ and $v_0 = 32$, then the position function is $h = -16t^2 + 32t + 48$.

Solution:

- Differentiating the position function and then equating the derivative to 0 gives the maximum height of the ball. Thus,

$$h = -16t^2 + 32t + 48$$

$$h' = -32t + 32 = 0$$

$$-32(t - 1) = 0$$

$$t = 1$$

Solution:

- Therefore, the ball will reach its maximum height after 1 second. To get the maximum height, substitute $x = 1$ in the original position function.

$$\begin{aligned} h &= -16t^2 + 32t + 48 = -16 + 32 + 48 \\ &= 64 \text{ feet} \end{aligned}$$

Relate Rates

- In this section, you will solve real-life problems concerning two or more related variables that change the function of time. These problems are commonly called *related-rates problems*, because their rate of change with respect to time are also related.

Relate Rates

- For example, you throw a stone in a pond. You will notice that the stone will cause ripples to form on the surface of the pond. The ripples change in size with respect to time – the radius of the ripples change, as well as its area. Also, the rate of change in the area of the ripples is directly related to the rate of increase in its radius.

Relate Rates

- Like optimization problems, related rates problems can be modelled and solved using differentiation. However, these problems do not require the use of explicit function. For example, you are given an equation involving variables x and y , and that both x and y are functions of the third variable t , which denotes time.

Relate Rates

- Because the rate of change of x with respect to time t and the rate of change of y with respect to time t are given, then you must differentiate both sides of the given equation with respect to time t by applying the chain rule. Hence, related-rates problems are solved using implicit differentiation.

In solving related-rates problems, several steps can be done:

- Draw a picture, or make a mathematical model of the situation.
- Label all quantities that may change as variables.
- Identify the independent variable.
- Identify the variables and their derivatives, what is being asked, and what are the given.

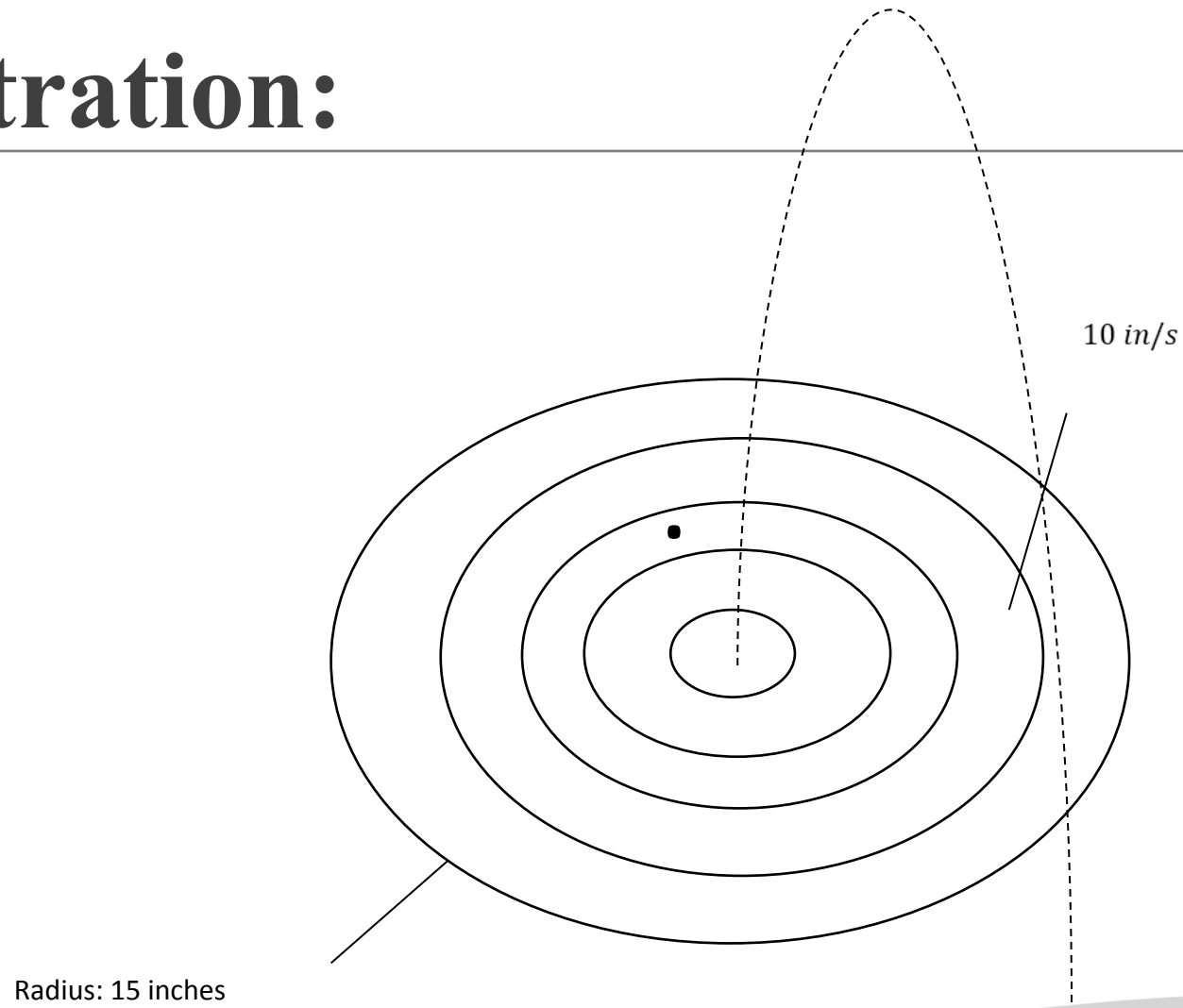
In solving related-rates problems, several steps can be done:

- Think of the relationship of the variables.
- Using the equation that relates the variables, differentiate with respect to time. Use implicit differentiation.
- Solve for the desired quantity.
- Go back over your work and write presentable solution.

Example 6:

- Eiron tosses a stone in a still pond, producing ripples shaped like concentric circles. One of the ripples has radius that increases constantly at 10 inches per second. At what rate is the area of the ripple increasing when the radius is 15 inches?

Illustration:



Solution:

- In a circle, the area is related to the radius by the equation $A = \pi r^2$. Let $\frac{dA}{dt}$ represent the rate of change of the area and $\frac{dr}{dt}$ the rate of change of the radius.
- Given: $\frac{dr}{dt} = 10$
- Asked: $\frac{dA}{dt}$ when $r = 15$

Solution:

- Use the equation for the area of the circle and differentiate with respect to time.

$$A = \pi r^2$$
$$\frac{d}{dt}[A] = \pi \frac{d}{dt}[r^2]$$
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Solution:

- To get $\frac{dA}{dt}$, we evaluate when $r = 15$ and $\frac{dr}{dt} = 10$:

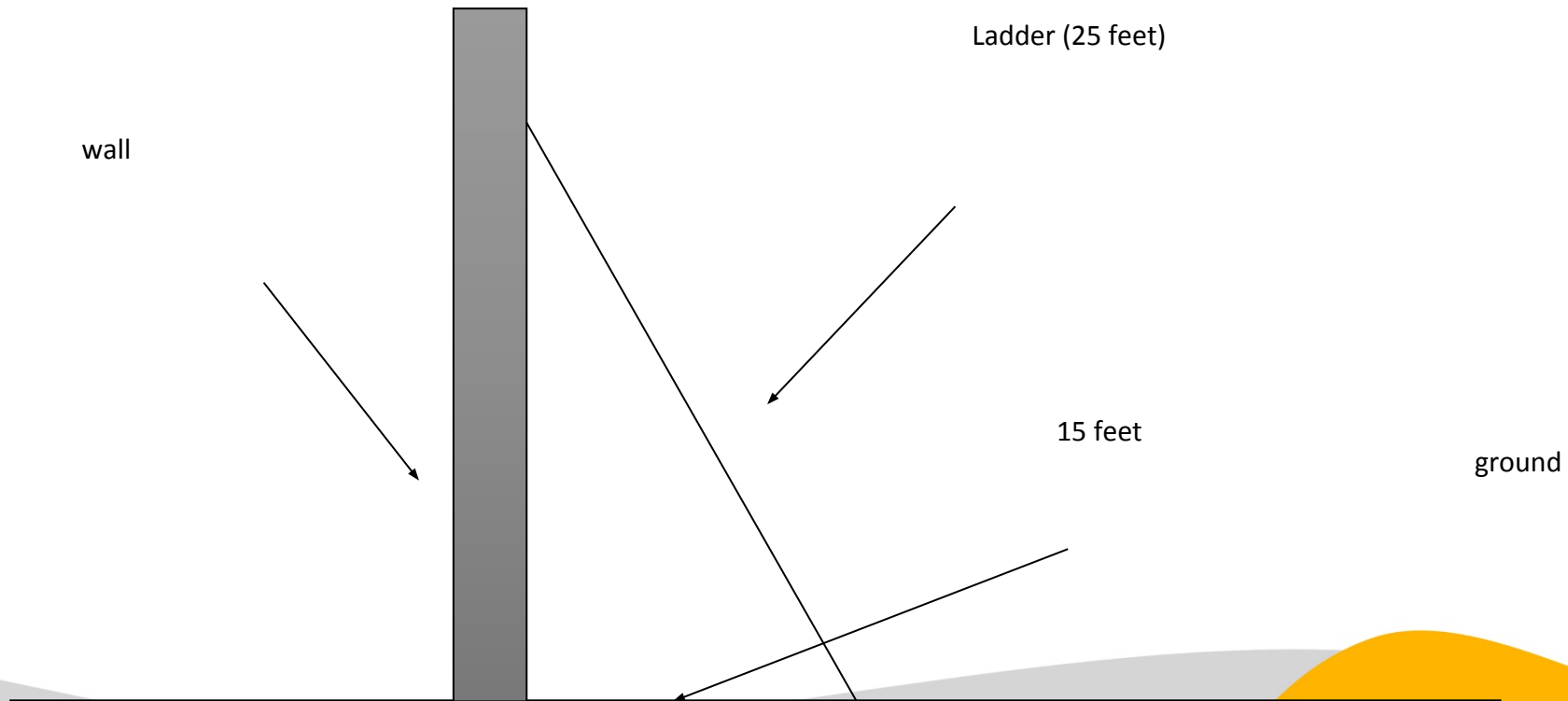
$$\begin{aligned}\frac{dA}{dt} &= 2\pi r \frac{dr}{dt} \\ \frac{dA}{dt} &= 2\pi(15)(10) \\ &= 300\pi\end{aligned}$$

- Therefore, the area of the circle is increasing at 300π square inches per second at the instant when the radius of the circle is 15 inches.

Example 7:

- A ladder 25 feet long is leaning against the vertical wall. If the bottom of the ladder is pulled horizontally away from the wall at 3 feet per second, how fast is the top of the ladder sliding down the wall, when the bottom is 15 feet from the wall?

Illustration:



Solution:

- Let t be the time (in seconds) that elapsed since the ladder started sliding down the wall, y be the distance (in feet) from the ground to the top of the wall, and x be the distance (in feet) from the bottom of the ladder to the wall. Because the bottom of the ladder is pulled horizontally away from the wall at 3 feet per second, $\frac{dx}{dt} = 3$. You wish to find $\frac{dy}{dt}$ when $x = 15$.

Solution:

- From Pythagorean theorem, you have

$$x^2 + y^2 = 25^2$$

$$y^2 = 625 - x^2$$

Solution:

- Because x and y are function of t , you must differentiate on both sides of the equation with respect to time t . You now obtain

$$2y \frac{dy}{dt} = -2x \frac{dx}{dt}$$
$$\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$$

Solution:

- If $x = 15$, then $y = 20$, because $y = \sqrt{625 - 15^2} = \sqrt{400} = 20$. Thus,

$$\frac{dy}{dt} = -\frac{15}{20}(3) = -\frac{9}{4}$$

- Therefore, the top of the ladder is sliding down the wall at the rate of $2\frac{1}{4}$ feet per second when the bottom is 15 feet from the wall. (The significance of the negative is that y is decreasing as x increasing.)

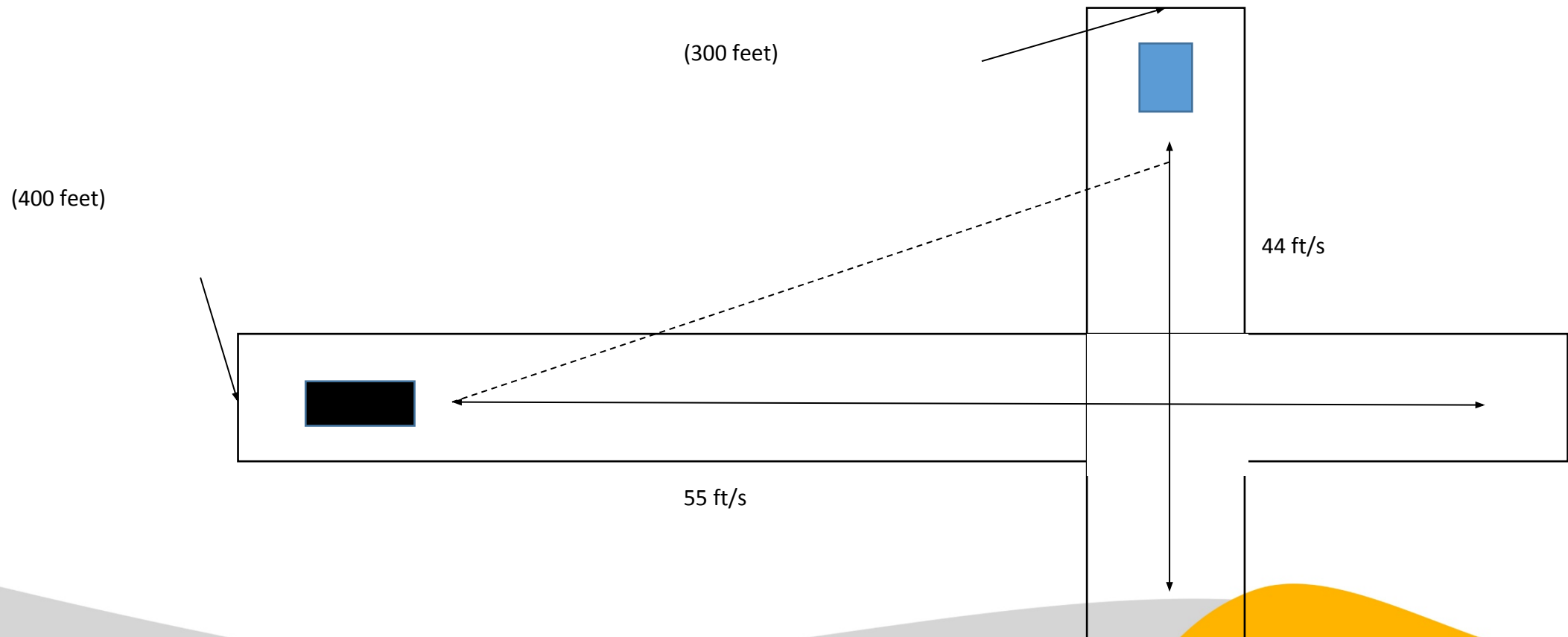
Relate Rates

- Although the details change from problem to problem, the general pattern of solutions is the same for all related-rates problems.

Example 8:

- Two cars, one going east at the rate of 55 feet per second, and the other going south at the rate of 44 feet per second, are traveling toward an intersection of their two roads. At what rate are the two cars approaching each other at the instant when the first car is 400 feet from the intersection and the second car is 300 feet from the intersection?

Illustration:



Solution:

- Make a representation of the problem, and assign the following.
- x = distance of the first car from the intersection
- y = distance of the second car from the intersection
- z = distance between the two cars at t seconds

Solution:

- First car is approaching the intersection at the rate of 55 feet per second, and because x is decreasing as t is increasing, then $\frac{dx}{dt} = -55$. Similarly, $\frac{dy}{dt} = -44$. You wish to find $\frac{dz}{dt}$ when $x = 400$ and $y = 300$.
- From the Pythagorean theorem, you have
$$x^2 + y^2 = z^2$$

Solution:

- Differentiating on both sides of the equation with respect to the time t , you obtain

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2z \frac{dz}{dt}$$

- This means that

$$\frac{dz}{dt} = \frac{2x \frac{dx}{dt} + 2y \frac{dy}{dt}}{2z}$$

Solution:

- Substituting $x = 400$ and $y = 300$ in the equation $x^2 + y^2 = z^2$ gives $z = 500$.

- $\frac{dz}{dt} = \frac{2x\frac{dx}{dt} + 2y\frac{dy}{dt}}{2z}$

- $= \frac{2(400)(-55) + 2(300)(-44)}{2(500)}$

- $= \frac{-44\ 000 - 26\ 400}{1\ 000}$

- $= \frac{-77\ 400}{1\ 000} = -77.4$

Solution:

- Therefore, the cars are approaching each other at the rate of 70.4 feet per second.

Example 9:

- A right circular cone has a radius that enlarges at a rate of 5 centimeters per second. The equation $h = 2r$ relates the radius and height of the cone. How fast is the volume of the cone changing at the instant when the radius is 8 centimeters?

Solution:

- The equation $V = \frac{1}{3}\pi r^2 h$ relates the volume, radius, and height of the right circular cone. In order to express the equation using only the volume and the radius, we substitute $h = 2r$.

Solution:

• Then,

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi r^2 (2r)$$

$$V = \frac{2}{3} \pi r^3$$

Solution:

- Given: $\frac{dr}{dt} = 5$
- Asked: $\frac{dV}{dt}$ when $r = 8$
- Differentiating the equation with respect to time t ,

$$V = \frac{2}{3}\pi r^3$$
$$\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{2}{3}\pi r^3\right]$$

Solution:

- $$\frac{dV}{dt} = \frac{2}{3} \pi (3r^2) \frac{dr}{dt}$$
$$\frac{dV}{dt} = 2\pi r^2 \frac{dr}{dt}$$
- To solve $\frac{dV}{dt}$, substitute $\frac{dr}{dt} = 5$ and $r = 8$.
$$\begin{aligned} \frac{dV}{dt} &= 2\pi r^2 \frac{dr}{dt} \\ &= 2\pi (8)^2 (5) \\ &= 640\pi \end{aligned}$$

Solution:

- So, when $r = 8$ cm, the volume of the cone is increasing at the rate of 640π (approximately 2010.62) cubic centimeters per second.